UTS Cloud Computing Class - Lab Guide

Week 5 Lab: vSwitches and Port groups

This week the lab tasks will involve configuring and setting up different Port Groups on existing vSwitches in the vSphere environment.

Task 1. Prepare for the lab.

Before powering ON the VMs, please check the Network Adaptor configured for all the VMs.

1. Open VMware Workstation, and **power on** the following six VMs: 'ESXi01', 'ESXi02', 'VCSA' and 'iSCSI' and 'OpenWRT'.
2. From the **console** of the '**DT**' (desktop) VM, open a **command prompt** and verify that you can **ping** each of the other machines. You should be able to ping ESXi01 (172.20.20.51) and ESXi02 (172.20.20.52), and VCSA (172.20.20.94).
   *** **Note:** Students using personal device should be able to run the ping command directly from the command prompt of the device.*
3. Connect to your VCSA vm using a **web browser** (https://172.20.20.94).
4. **Verify** that your hosts are connected to the iSCSI target from last week.
5. **Verify** that there is already a vm on each host from earlier weeks.

* **STOP** * *If connectivity is not established follow the steps outlined from Week 2 Task 3 before continuing.*

Task 2. Create VM using OVF file

In the earlier weeks, the VM within the ESXi hosts is created using the '.iso' file that is attached to the respective Datastores. The VMs were booting from the '.iso' files and they were not installing the OS into the VM. Upon VM migration, the VM would not boot on the other host and may give 'Operating System Not Found' error.
In order to create a VM with OS installed on it, this another method to create VM using the OVF/OVA file is used.

1. From the web browser for ESXi01 (https://172.20.20.51), on left side panel '**Virtual Machines**' option, click **Create/Register VM.**
2. In this wizard, select the second option to create a vm **Deploy a virtual machine from an OVF or OVA file,** click **Next**.
3. Enter a name similar to the previous vm, *YourName*-vm3 (say) and click on the given box to select the OVF files. You might see the directory **TinyCore+Script_v2.0,** enter into the directory and select all the files and click **Open.**
4. Select the storage as **Datastore1**, click **Next.**
5. Leave the rest of the settings as default, click **Next** and click **Finish**. The newly created vm will appear in the list of virtual machines.
6. Repeat steps 1-5 for ESXi02 (https://172.20.20.52) to create a vm using OVF file.

Task 3. Test VM connectivity.

We must ensure that connectivity between VMs is currently established to test future tasks. **** Note:*** *These vms will not be able to access mouse when using DT. The keystrokes to access the vm are given at the end of the document. The online students will be able to access the mouse when the vms are hosted on VMware Workstation. The steps are shown in the additional document in Week 5 Lab module.*

1. **Clone** the vm created using OVF method so that you will have two vms.
2. To **clone** a vm, on the VCSA we browser, right click on the vm to be cloned and select the option **Clone -> Clone to Virtual Machine**.
3. Give name to the virtual machine and click **Next.**
4. Select the appropriate ESXi host and click **Next.**
5. Select **Datastore** as storage and under clone options select **Power on virtual machine after creation** option and click **Next** and **Finish.**
6. Once you have two VMs running on the same host, you can test connectivity between them.
7. Make sure the two VMs are running and make sure they can **ping** each other. **** Note:*** *Check the IP for both the vms and configure the IPs in the same subnet. To check the IP, use 'ifconfig' command on TinyCore Linux vm.*
8. What is the IP of your first VM? ______________________
9. What is the IP of your second VM? ____________________ 6. What is the name of the Network that the VMs are connected to?

Task 4. Change your VMs network connectivity.

We are going to modify the network of just one of our two VMs to affect our connectivity. We will do this by placing the VM into a different VLAN which will restrict their ability to communicate. (Only machines in the same VLAN can communicate).

1. On the VCSA web browser, select any ESXi host.
2. Go to **Configure tab -> Networking -> Virtual switches**.
3. On this page click **Add Networking,** select **Virtual Machine Port Group for Standard Switch** option and click **Next**.
4. Select the existing standard switch **vSwitch0.**
5. Name the network label as **'Production'** and assign a VLAN ID as **100** to the portgroup.
6. **Edit settings** on your second VM and modify the network adapter so that it is now connected to the **'Production'** portgroup and not 'VM Network' which is the default.
7. Once verified, ping the first VM.
8. Was the test successful? _______________________
9. Why? ______________________

Task 5. vMotion your VMs to your other host.

Currently you should have 2 VMs both running on any 1 host, and you will attempt vMotion to move both of them to the other host in your environment.

1. **Migrate** your first VM to your other host.
2. Was the migration successful? Why/Why not? _________________________________

3. **Migrate** your second VM to your other host.
4. Was the migration successful? Why/Why not? ___________________________________
5. Without changing the network of the affected VM, ensure that the **vMotion** is successful between hosts. This is a troubleshooting exercise. Once resolved, you will have learned one of the necessary requirements for a vMotion to be performed.

Task 6. Restore Connectivity between VMs

To ensure that the VMs can communicate, they need to be on the same network, but not necessarily in the same portgroup.

1. Create a new portgroup on both hosts called **'Dev'**. Do not assign a VLAN ID.
2. Connect the second VM (the one in the Production portgroup) to the 'Dev' portgroup instead.
3. You should have one VM in 'VM Network' and one VM in 'Dev'.
4. Conduct ping tests from the VMs.
5. Were the pings successful? Why/Why not? ___________________________________

Task 7. Clean up for next lesson

Shutdown your VMs in the web client, then shutdown all your VMs in VMware Workstation.

Keys to access through the vm:

- Right mouse button on the vm window: Pop-up menu
- Up and Down arrow keys: To browser the menu
- Right and Left arrow key: To expand the menu
- SPACE BAR key: To select the highlighted menu option
- TAB key (ControlPanel-> Network): To navigate to various options in the open window • ESC key: To close the current working window